MODEL BENCHMARK · BONUS WORK, AFTER THE TIMEBOX

Three models, two reasoning efforts, two real assessments

5 November 2026 · Opus 5.5, Sonnet 5.5 and Haiku 5.5, each at effort **medium** and **extra high** (xhigh) · 24 reports, two per configuration and assessment, through Nettle's normal queue and worker · $8.36 in total, judges included

Set-up

- **Pacific Coast** (Default Insurer): COPE survey in English from 3 photos and 2 notes. Thin evidence. **Udine** (Customer A): COPE survey in Italian from a 4-page survey PDF and 3 photos.
- Two Opus judges. **Recall**: are the five recommendations in the Udine PDF covered? **Support**: do the quotes behind each rated answer justify the option chosen over the others?

Results (times and costs: mean of two runs; other columns: totals over both runs and both assessments)

Model	Effort	Pacific Coast time · cost	Udine time · cost	Quotes verified	Expert recs (Udine)	Rated answers supported	Not supported	Rejected, retried
Opus 5.5	medium	107 s · $0.26	169 s · $0.47	244 / 244	5 / 5	10 / 22	1	0
	xhigh	220 s · $0.52	484 s · $1.31	386 / 386	5 / 5	9 / 24	3	0
Sonnet 5.5	medium	52 s · $0.10	87 s · $0.19	171 / 171	5 / 5	6 / 21	0	1
	xhigh	135 s · $0.22	274 s · $0.50	262 / 262	5 / 5	8 / 22	1	0
Haiku 5.5	medium	58 s · $0.010	118 s · $0.017	152 / 152	4.75 / 5	5 / 24	4	2
	xhigh	214 s · $0.026	372 s · $0.051	231 / 231	5 / 5	7 / 23	3	0

Findings

- **The mechanics hold everywhere.** All 1,446 quotes were found verbatim in the evidence, and the expert's recommendations were covered. The one 4.5 out of 5 is judge noise: the same report scored 5 out of 5 in another pass.
- **The model matters more than the effort.** At medium, Opus's ratings are the best supported (10 of 22, against 6 of 21 and 5 of 24). Haiku overreaches: in all three runs it rated sprinkler coverage "Partial" from a quote about corrosion. **A verified quote proves the evidence exists, not that it supports the answer.**
- **Extra high costs 2 to 3 times more and takes 2 to 4 times longer** (Opus: 8 minutes and $1.31 for Udine). It cites 55 to 75% more evidence and removes format slips: no rejected answers in 12 runs, against 3 at medium.
- **It doesn't make ratings better supported.** Unsupported ratings rise from 5 to 7, and Opus drops from 45% to 38% supported. At extra high, Opus rated sprinkler coverage "Full" once, where at medium it had declined both times.
- **The template has a weak question.** Every model, at both efforts, rated Udine's external exposure (from neighbours) with quotes about earthquakes and floods, which another question covers. A guidance fix, found by the support check.

Recommendation

- **Opus 5.5 at medium** by default for reports that will be issued: the best-supported ratings, at a fraction of extra high's cost and time.
- **Don't raise effort by default.** Sonnet at extra high costs about as much as Opus at medium, but is slower and less well supported. Sonnet at medium suits cost-sensitive customers and drafts; Haiku suits previews and triage.
- **Fix the external exposure question,** then run the support check in the product to flag the rated answers the quotes don't justify, before an engineer's review.

Limits

- Two runs per configuration, and judges that are noisy and strict ("partial" is their most common verdict), so differences of one to three answers are noise. Recall saturates because the advice is in the evidence: hold it out next time.

Full write-up: docs/benchmark.md · every verdict with its reason: docs/benchmark-results.json
Reproduce: `python manage.py benchmark_models --efforts medium xhigh --support-check` (about $8.50 and an hour with two workers)